

LAB EXPERIMENT RECORD

Set 5: Website Analytics

Department	AIML
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze website traffic data using a line chart, bar chart, and stacked area chart, and to build an interactive dashboard using R and Shiny.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

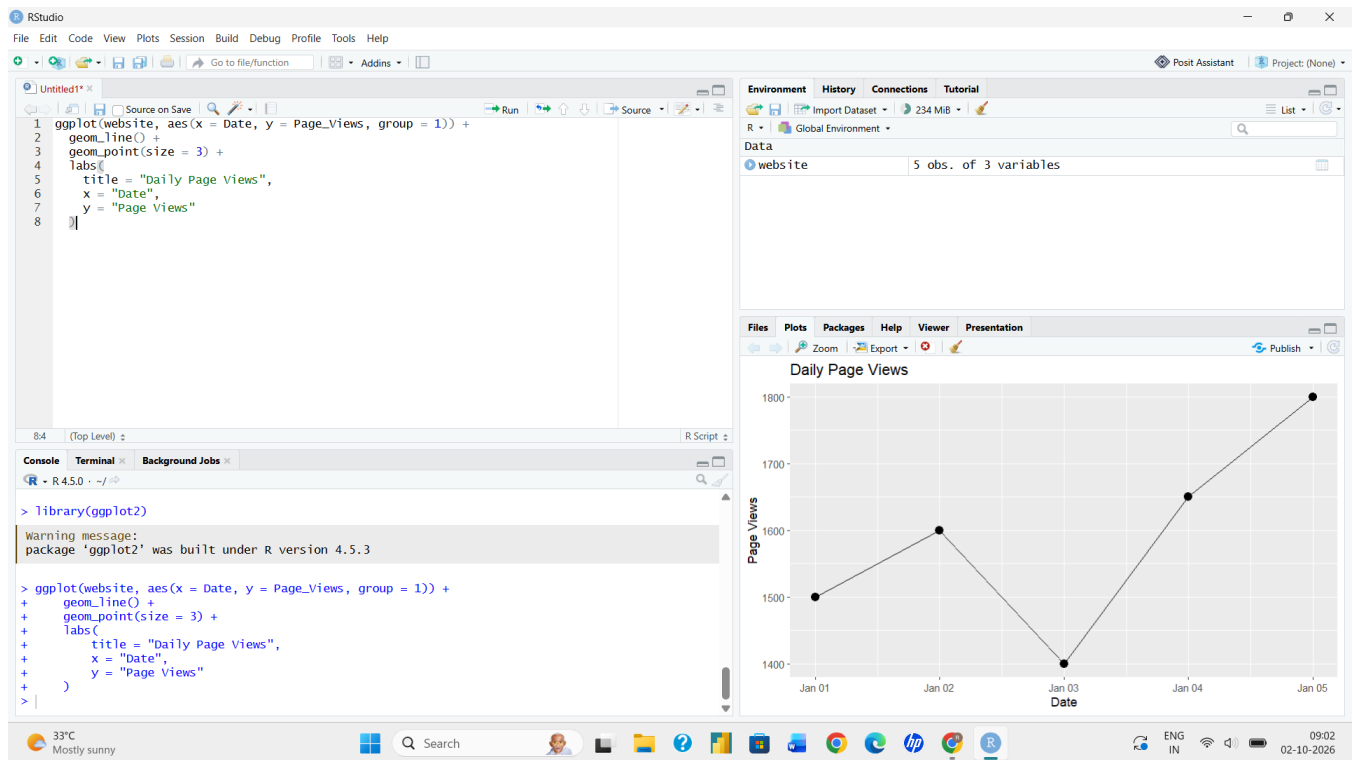
Note: The given dataset does not contain user interaction data. For Program 3 the daily likes, shares and comments are assumed values (Likes: 120, 150, 100, 160, 180; Shares: 40, 55, 30, 50, 65; Comments: 20, 25, 15, 30, 35).

PROGRAM 1: Line Chart of Daily Page Views

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))
views <- c(1500, 1600, 1400, 1650, 1800)

# Line Chart
plot(date, views,
     type = "o",
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views",
     col = "blue",
     pch = 19)
```



Output 1: Daily Page Views Line Chart

Observation: Page views rise from 1500 on 1 January to 1600 on 2 January, fall to the lowest value (1400) on 3 January, and then increase steadily to the highest value (1800) on 5 January.

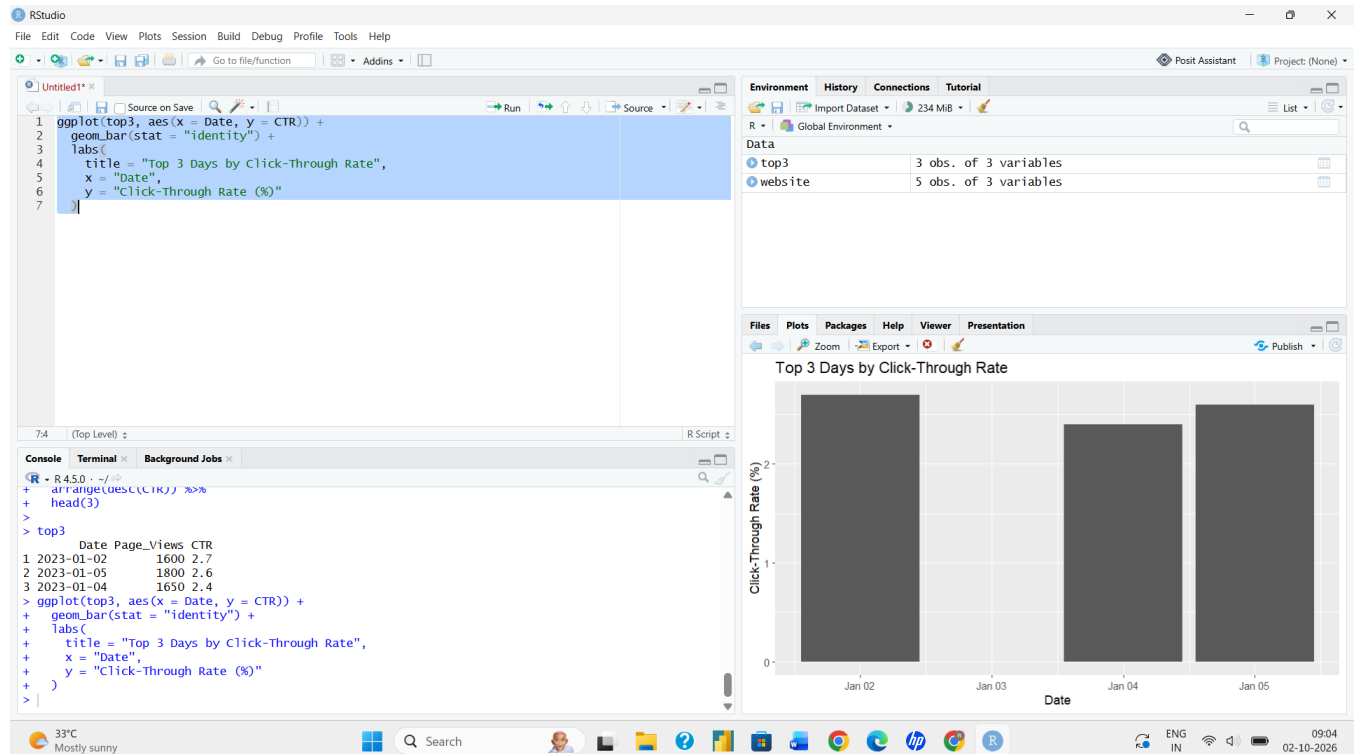
PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

Generate a bar chart showing the top N days with the highest click-through rates (here N = 3). Label the chart elements.

```
# Dataset
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                 "2023-01-04", "2023-01-05"))
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# Top N days with the highest click-through rate
N <- 3
top <- order(ctr, decreasing = TRUE)[1:N]

# Bar Chart
barplot(ctr[top],
        names.arg = format(date[top], "%Y-%m-%d"),
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = paste("Top", N, "Days by Click-through Rate"),
        col = "orange",
        ylim = c(0, 3))
```



Output 2: Top 3 Days by Click-through Rate

Observation: The three days with the highest click-through rates are 2 January (2.7%), 5 January (2.6%) and 4 January (2.4%). Changing the value of N in the program changes how many days are shown.

PROGRAM 3: Stacked Area Chart of User Interactions

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

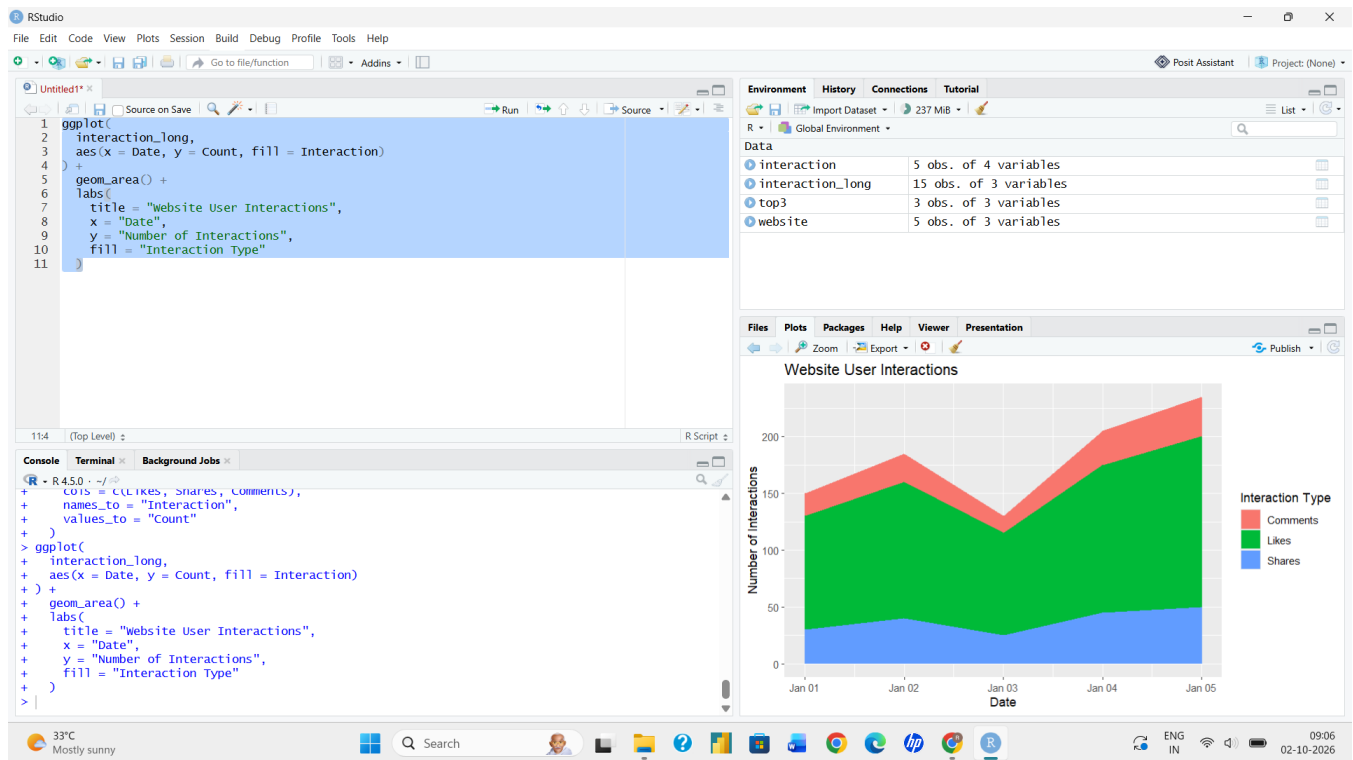
```
# Dataset (user interaction values assumed)
date  <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))
likes <- c(120, 150, 100, 160, 180)
shares <- c(40, 55, 30, 50, 65)
comments <- c(20, 25, 15, 30, 35)

# Cumulative values for stacking
c1 <- likes
c2 <- likes + shares
c3 <- likes + shares + comments
x <- 1:5

# Stacked Area Chart
plot(x, c3,
     type = "n",
     xaxt = "n",
     xlab = "Date",
     ylab = "Number of Interactions",
     main = "Distribution of User Interactions",
     ylim = c(0, max(c3) + 20))

polygon(c(x, rev(x)), c(c3, rev(c2)), col = "seagreen", border = NA)
polygon(c(x, rev(x)), c(c2, rev(c1)), col = "orange", border = NA)
polygon(c(x, rev(x)), c(c1, rep(0, 5)), col = "steelblue", border = NA)

axis(1, at = x, labels = format(date, "%b %d"))
legend("topleft",
     legend = c("Comments", "Shares", "Likes"),
     fill = c("seagreen", "orange", "steelblue"))
```



Output 3: Distribution of User Interactions

Observation: Likes form the largest share of interactions on every day, followed by shares and then comments. The total number of interactions is highest on 5 January (280) and lowest on 3 January (145), following the same pattern as the page views.

PROGRAM 4: Interactive Website Traffic Dashboard

Build a dashboard with the line chart and bar chart for interactive exploration of website traffic data using the Shiny package.

```
library(shiny)

date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                 "2023-01-04", "2023-01-05"))
views <- c(1500, 1600, 1400, 1650, 1800)
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

ui <- fluidPage(

  titlePanel("Website Traffic Dashboard"),

  sidebarLayout(

    sidebarPanel(
      selectInput("chart",
                 "Select Chart:",
                 choices = c("Line Chart", "Bar Chart")),

      sliderInput("n",
                 "Top N days (Bar Chart):",
                 min = 1, max = 5, value = 3, step = 1)
    ),

    mainPanel(
      plotOutput("trafficPlot")
    )
  )
)

server <- function(input, output) {

  output$trafficPlot <- renderPlot({

    if (input$chart == "Line Chart") {

      plot(date, views,
           type = "o",
           xlab = "Date",
           ylab = "Page Views",
           main = "Daily Page Views",
           col = "blue",
           pch = 19)

    } else {

      top <- order(ctr, decreasing = TRUE)[1:input$n]

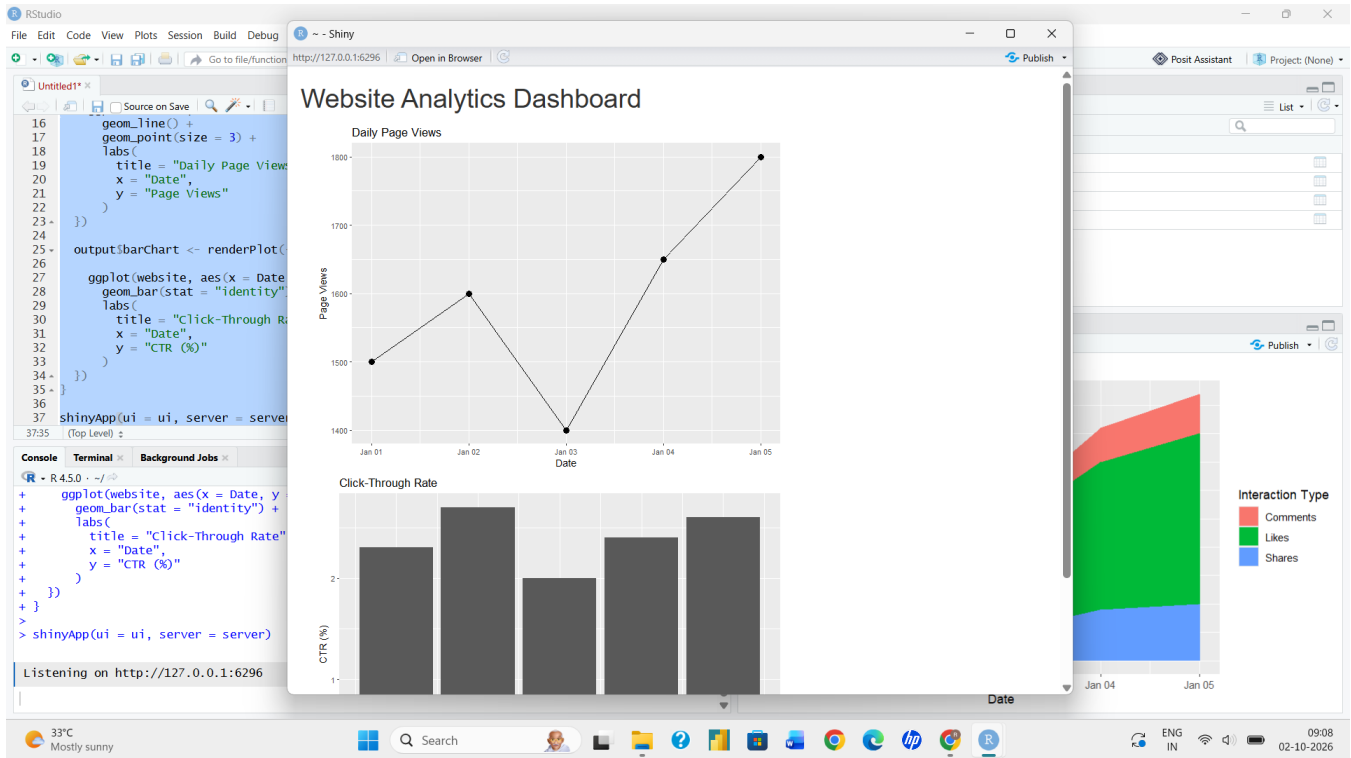
      barplot(ctr[top],
              names.arg = format(date[top], "%Y-%m-%d"),
              xlab = "Date",
              ylab = "Click-through Rate (%)",
              main = paste("Top", input$n, "Days by Click-through Rate"),
              col = "orange")
    }
  })
}
```

```

}
})
}

shinyApp(ui = ui, server = server)

```



Output 4: Interactive Website Traffic Dashboard

Observation: The dashboard provides a chart-selection control and a slider for N. Selecting Line Chart displays the daily page-view trend, while selecting Bar Chart displays the top N days by click-through rate.

Procedure

1. Create the date, page views and click-through rate vectors in R.
2. Use `plot()` with `type = "o"` to create the line chart of daily page views.
3. Use `order()` to find the top N click-through rates and `barplot()` to display them.
4. Create the likes, shares and comments vectors, calculate cumulative totals, and use `polygon()` to draw the stacked area chart.
5. Install and load the Shiny package, then create the UI and server components for the interactive dashboard.
6. Run the Shiny application, switch between the Line Chart and Bar Chart, and change N using the slider.

Result

The website traffic data was successfully visualized using a line chart, bar chart, and stacked area chart. An interactive Shiny dashboard was also created to explore the page-view trend and the top N days by click-through rate.

Key R Functions Used

Function	Purpose
<code>as.Date()</code>	Converts text values into dates.
<code>plot()</code>	Creates line charts and the base of the area chart.
<code>order()</code>	Sorts values to find the top N days.
<code>barplot()</code>	Creates bar charts.
<code>polygon()</code>	Draws the filled areas of the stacked area chart.
<code>legend()</code>	Adds a legend to the chart.
<code>fluidPage()</code>	Creates the Shiny page layout.
<code>selectInput()</code> / <code>sliderInput()</code>	Provide the chart selection and N controls.
<code>renderPlot()</code>	Generates plots dynamically in Shiny.
<code>shinyApp()</code>	Runs the interactive Shiny application.